CONTACT INFORMATION

JACKET DEMBY'S

Research Interests

- Machine Learning
- Deep Learning
- Robotics
- Assistive Technologies

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EDUCATION

Doctor of Philosophy (Ph.D.)
Still attending
GPA: 3.724/4

University of Missouri, Columbia, Missouri, USA
Electrical and Computer Engineering
Expected Graduation: December 2024

Master of Science (MS)
Graduated July 2020
GPA: 3.696/4

University of Missouri, Columbia, Missouri, USA
Computer Engineering

Engineering Diploma
Graduated July 2015
GPA: 15.06/20

Ucac-Icam Institute, Douala, Cameroon
Generalist Engineering
Top of class

MOST RECENT WORK EXPERIENCE

Display Electrical Engineer Intern | May 2023 – August 2023
Apple – Cupertino, California, United States of America

- Utilized state-of-the-art super resolution deep learning methods to improve display technology.
- Explored alternative hyper parameters and evaluation metrics tailored for display technology.

Skills: PyTorch · Data Collection · Data Preparation · Super Image Resolution · Applied Research · Interpersonal Communication · Presentation Skills · MATLAB · Python (Programming Language) · Deep Learning

Junior Project Engineer | December 2016 – July 2017
Total E&P Congo - Pointe-Noire, Republic of Congo

- Prepared, supervised and generated daily reports for construction activities of the Floating Production Unit (FPU) Likouf alongside contracted companies.
- Worked on Pressure Safety Valves (PSV) Recalibration Campaign, managed a database of the PSV joint stocks, and ensured team members compliance with on-site safety measures.

Maintenance Engineer | September 2015 – October 2016
Sogea-Satom - Brazzaville, Republic of Congo

- Monitored the computer-assisted supply chain and maintenance management software and produced monthly analysis reports.
- Developed and maintained Excel VBA (Visual Basic for Applications) programs for monthly stocks' management analysis, optimization and periodic inventories follow up.

Intern Maintenance Engineer by Alternance | January 2013 – December 2014
Sogea-Satom - Brazzaville, Republic of Congo

- Monitored the computer-assisted supply chain and maintenance management software.
- Continually improved methods and procedures for processes, stocks' management, documenting and workflow techniques.

SOME COURSEWORK

- Introduction to Mechatronic and Robotic Vision
- Probability and Stochastic Processes for Engineers
- Introduction to Machine Learning and Pattern Recognition
- Parallel Programming for High Performance Computing
- Introduction to Computational Intelligence
- Digital Image Processing
- Computer Vision
- Supervised Learning
- Neural Networks and Deep Learning
- Real-Time Embedded Systems

TEACHING EXPERIENCE

Teaching Assistant, University of Missouri-Columbia, Columbia, Missouri, United-States

- **Spring 2020 : Neural Networks**
- **Fall 2020 / 2021 / 2022 / 2023 : Building Intelligent Vision-Guided Robots**
- **Spring 2021 : Introduction to Machine Learning and Pattern Recognition**
- **Spring 2022 : Power Electronics I**
- **Spring 2023 : Architectural Robotics**

RESEARCH EXPERIENCE

August 2017 – Present: Research Assistant (Vision Guided and Intelligent Robotics Lab – ViGIR Lab)

- Behavior-based mobile robotics using multiple sensors (RGB, Intelsense, Sonars, Velodyne)
- Accuracies of approximate solutions provided by analytical, data-driven (deep learning-based), numerical, and hybrid inverse kinematics solvers for robotic manipulators
- Choosing the correct generalized inverse for the numerical solution of the inverse kinematics of incommensurate robotic manipulators
- Object detection and pose estimation based on stereo vision and deep learning algorithms in resource-limited embedded hardware

TECHNOLOGIES

Operating Systems: Linux, Windows, MacOS

Programming languages: Python, C/C++, MATLAB, Arduino, Processing, Cuda, Lyx/Latex

Software and libraries:

- **Robotics**: ROS (Robot Operating System)
- **Machine learning**: Keras, Pytorch, TensorFlow
- **Computer vision**: OpenCV, PCL
- **Mechanical**: Solidworks, Ansys Fluent

PUBLICATIONS

1. **J. Demby's**, Y. Gao and G. N. DeSouza, "A Study on Solving the Inverse Kinematics of Serial Robots using Artificial Neural Network and Fuzzy Neural Network," *in 2019 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE 2019)*, New Orleans, LA, USA, 2019, pp. 1-6. *(**Accepted**)*.

2. **J. Demby's**, Y. Gao, A. Shafiekhani, G.N. DeSouza, "Object Detection and Pose Estimation Using CNN in Embedded Hardware for Assistive Technology, " *in 2019 IEEE Symposium Series on Computational Intelligence (IEEE SSCI 2019)*, December 2019. *(**Accepted**)*.

3. **J. Demby's**, A. Shafiekhani, F.B Fritschi and G. N. DeSouza, "Spatio-Temporal Reconstruction and Visualization of Plant Growth for Phenotyping," *in 2021 IEEE Symposium Series on Computational Intelligence (IEEE SSCI 2021)*, December 2021. *(**Accepted**)*.

4. **J. Demby's**, J. Uhlmann and G. N. DeSouza, "Achieving Unit-Consistent Pseudo-Inverse-based Path-Planning for Redundant Incommensurate Robotic Manipulators". arXiv preprint arXiv:2308.02954. *(**Journal paper in preparation for resubmission**)*.

5. **J. Demby's**, J. Uhlmann and G. N. DeSouza, "Choosing the Correct Generalized Inverse for the Numerical Solution of the Inverse Kinematics of Incommensurate Robotic Manipulators". arXiv preprint arXiv:2308.02964. (***Journal paper in preparation for resubmission***)

ACHIEVEMENTS AND ACTIVITIES

May 2019 – present	Student Member of IEEE.
December 2017	Best 3rd final project in Digital Image Processing class (Fall 2017).
August 2017 – present	Member of the University of Missouri Fulbright Organization (**MUFO**).
July 2017 – present	Recipient of the **Fulbright Scholarship** in Congo Brazzaville to pursue a M.Sc. in Computer Engineering at the University of Missouri-Columbia.
January 2017	Participant for the Business and Entrepreneurship Track of the Young African Leaders Initiative (YALI Program) in Nairobi, Kenya.
April 2016	2nd prize ($13,000) of the 2015-2016 national Startup Challenge of Total E&P in Congo Brazzaville.
January 2015 – June 2015	Participant of the internal Icam exchange program to conduct an industrial mission on Computational Fluid Dynamics (CFD) in Lille, France.
June 2012 – July 2013	Recipient of the **Sogea-Satom scholarship** to study at Ucac-Icam Cameroon.
August 2010 – June 2012	Recipient of the **Total E&P scholarship** to study at Ucac-Icam Congo.

SOME COURSEWORK PROJECTS INVOLVING FINAL PRESENTATIONS

Class: Real-Time Embedded Systems (Spring 2020)
Project: Data generation for supervisory control and data acquisition system (DataGen-SCADA)
Main objective, results and tools: Build a real-time data generation simulator for distributed SCADA systems using IoT devices; to gather and analyze data in real-time industrial settings and behaviors. Python, Processing, Arduino and ESP8266 devices were used.

Class: Neural Networks (Spring 2019)
Project: Solving the inverse kinematics of a 4 DoF SCARA robot using extreme learning machines and radial basis function networks
Main objective, results and tools: Conduct a comparative study between extreme learning machines and radial basis function networks in solving the inverse kinematics of the chosen robotic arm. Lowest MSE (Mean-Squared Error) results were obtained with extreme learning machines. Matlab and Python were used.

Class: Introduction to Computational Intelligence (Fall 2018)
Project: Solving the inverse kinematics of a 6 DoF Kinova robot using artificial and fuzzy neural networks
Main objective, results and tools: Conduct a comparative study between artificial and fuzzy neural networks in solving the inverse kinematics of the chosen robotic arm. A trade-off was observed in terms of precision of the predicted joint configuration depending on the workspace. Matlab and Python were used.

Class: Computer Vision (Spring 2018)
Project: Hardware integration of a real time object detection CNN (Yolo V2)
Main objective, results and tools: Develop an android mobile application based on YOLO V2 to recognize and pronounce objects in real-time. Android, Darknet and a pretrained Yolo V2 model were used.

Class: Introduction to Mechatronic and Robotic Vision (Fall 2017)
Project: Investigating the control of a robotic arm with a real-time tracking of human hand motions
Main objective, results and tools: Track human hand motions using a Kinect sensor and control the robotic arm with the hand motions tracking information. C/C++ and ROS were used.

Class: Digital Image Processing (Fall 2017)
Project: Super Image Restoration (SIR)
Main objective, results and tools: Develop a Matlab application with a friendly Graphical User Interface (GUI) to restore old damaged images with cracks in only three click steps. Matlab, Python and Algorithmia Colorization API were used.

Languages and Interests

Languages

- French: native language
- English: fluent (speaking, reading, writing)

Interests

Sport: Walking, Running, Volleyball, Swimming
Board games: Scrabble, Ludo, Chess, Checkers

Volunteer Work

Intervarsity at University of Missouri-Columbia | Volunteer photographer and events' organizer

- Organize events (games, discoveries, English clubs, shopping, etc.) to help international students feel welcome at the university.
- Take pictures during the events and share them with all the members.

University of Missouri Fulbright Organization (MUFO) | Volunteer

- Work with the International Center to provide rides to new international sponsored students.
- Lead scavenger hunt activities to help international sponsored students discover and locate key facilities and resources on campus.

OVA (Œuvre de Vie et d'Amour) Association in Cameroon | Volunteer

- Organize activities on the International Day of the African Child and Christmas for street children and children with special needs.